

Lecture 27-SYSLOG (System Logging Protocol)

1 What is SYSLOG?

SYSLOG (System Logging Protocol) is an **industry-standard protocol** used for **message logging on network devices**.

Network devices such as:

- Routers
- Switches
- Firewalls
- Servers

generate **log messages** when events occur.

 These messages are called **logs**.

SYSLOG records:

- System events
- Errors
- Warnings
- Status changes

2 Why SYSLOG is Needed?

Imagine this situation:

At **2:00 AM**, a company router suddenly stops working. 🌙

Without logs ❌

- Engineers don't know what happened
- Troubleshooting becomes very difficult

With SYSLOG 

The logs will show exactly what happened:

- Interface went down
- OSPF neighbor lost
- Device restarted
- Authentication failed

 Think of SYSLOG like a **CCTV camera for network devices**.

It records everything happening in the system.

3 What Events Does SYSLOG Log?

SYSLOG records many system events such as:

- ✓ Interface UP or DOWN
- ✓ OSPF neighbor UP or DOWN
- ✓ System reboot or restart
- ✓ Configuration changes
- ✓ Login attempts
- ✓ Security violations
- ✓ Hardware errors

These logs help engineers **troubleshoot problems quickly**.

4 Where SYSLOG Messages Can Be Stored?

SYSLOG messages can be stored in several locations.

Location	Description
CLI Console	Displayed directly in the device console
Buffer (RAM)	Stored temporarily in device memory
VTY Lines	Displayed during Telnet or SSH sessions

Location	Description
	External Syslog Server Sent to a centralized logging server

Logs are extremely important for:

- Troubleshooting issues
- Security monitoring
- Auditing network activity

5 **SYSLOG and SNMP Relationship**

SYSLOG and SNMP are both used for **network monitoring and troubleshooting**, but they work differently.

SYSLOG

Logs events

Push-based

Records system messages

No configuration control

SNMP

Monitors device data

Pull-based

Retrieves device information

Can modify variables

 Think of it this way:

- **SYSLOG → Event recorder** 
- **SNMP → Monitoring system** 

Both work together in real networks.

6 **SYSLOG Message Format (VERY IMPORTANT)**

SYSLOG messages follow a specific format:

seq: timestamp: %facility-severity-MNEMONIC: description

Some fields may not appear depending on device configuration.



7 Explanation of Message Fields

Field	Meaning
seq	Sequence number of message
timestamp	Time when message was generated
facility	Process that generated the message
severity	Importance level of the event
mnemonic	Short code describing the event
description	Detailed explanation



8 What is a Mnemonic?

A **Mnemonic** is a short code that indicates the event.

Example:

LINK-3-UPDOWN

Meaning:

Part	Meaning
LINK	Interface event
3	Severity level
UPDOWN	Interface status changed



SYSLOG Severity Levels (VERY IMPORTANT)

SYSLOG uses **8 severity levels** ranging from **0 to 7**.

These levels indicate **how serious the event is**.

Level	Keyword	Meaning
0	Emergency	System unusable
1	Alert	Immediate action required
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notification	Normal but significant
6	Informational	Informational messages
7	Debugging	Debug messages



Easy Memory Trick

To remember the severity levels, use this mnemonic:

Every Awesome Cisco Engineer Will Need Ice Cream Daily 🍦

Letter Level

E	Emergency
A	Alert
C	Critical
E	Error
W	Warning
N	Notification

Letter Level

I Informational

D Debugging



Importance in Real Companies

SYSLOG severity levels are used to define **incident priority**.

Example in IT operations:

Concept Meaning

SLA Service Level Agreement

MTTR Mean Time To Resolve

MTR Mean Time To Respond

ITIL Incident management framework

Severity level helps determine:

- Issue priority
 - Response time
 - Escalation level
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SYSLOG Logging Locations



1. Console Line

- Logs appear in console CLI
 - Used during direct device access
 - Default level: **0–7**
-

2. Buffer (RAM)

Logs stored in device memory.

Important:

- Logs are **lost after reboot**
 - Useful for quick troubleshooting
-

3. VTY Lines (Telnet / SSH)

Logs displayed during remote access.

Important:

- Disabled by default
 - Requires configuration
-

4. External Syslog Server

Most common method in real networks.

Advantages:

- Centralized logging
- Long-term storage
- Used for security audits

 Syslog servers listen on:

UDP Port 514

1 1 Severity Level Behavior

When configuring logging level:

Example:

logging trap warning

Router will log:

- Warning (4)
- Error (3)
- Critical (2)
- Alert (1)
- Emergency (0)

 Logging always works **upward toward level 0**.

1 2 Terminal Monitor (IMPORTANT)

When connected through **Telnet or SSH**, syslog messages are **not displayed by default**.

To enable them:

R1# terminal monitor

Important:

You must run this command **every time you reconnect**.

1 3 Logging Synchronous

Problem:

If a syslog message appears while you are typing a command, it can interrupt the command line.

Solution:

logging synchronous

This command ensures the CLI remains **clean and readable**.

Service Timestamp and Sequence Numbers

These commands improve log readability.

Enable timestamps:

```
service timestamps log datetime msec
```

Enable sequence numbers:

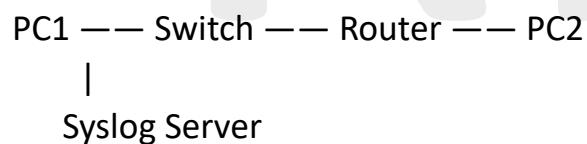
```
service sequence-numbers
```

These help identify:

- Exact time of events
- Order of events

SYSLOG LAB (Packet Tracer)

Lab Topology



IP Addressing

Device	IP
PC1	192.168.1.12
Server	192.168.1.100
Router G0/0	192.168.1.1

Router Credentials

Username: jeremy

Password: ccna

Enable password: ccna

TASK 1 — Console Syslog

Step 1

Open console connection:

PC2 → Desktop → Terminal

Step 2 — Shutdown Interface

R1> enable

R1# configure terminal

R1(config)# interface g0/0

R1(config-if)# shutdown

A syslog message will appear.

Step 3 — Enable Interface

R1(config-if)# no shutdown

Severity Level

Interface shutdown usually generates:

Severity Level 3 (Error)

Step 4 — Enable Timestamp

R1(config)# service timestamps log datetime msec

TASK 2 — Telnet Logging

Step 1 — Connect via Telnet

From PC1:

```
telnet 192.168.1.1
```

Step 2 — Enable Interface

```
R1(config)# interface g0/1
```

```
R1(config-if)# no shutdown
```

No syslog message appears.

Why?

Because logging to VTY is disabled.

Step 3 — Enable Terminal Monitor

```
R1# terminal monitor
```

Now syslog messages will appear.

TASK 3 — Logging to Buffer

Enable buffer logging:

```
R1(config)# logging buffered 8192
```

Verify logs:

```
R1# show logging
```

TASK 4 — External Syslog Server

Step 1 — Configure Server

```
R1(config)# logging host 192.168.1.100
```

```
R1(config)# logging trap debugging
```

Step 2 — Generate Event

```
R1(config)# interface g0/1  
R1(config-if)# shutdown
```

Step 3 — Verify on Server

Go to:

Server → Services → Syslog

You should see the received log messages.

Real-World Example

Example: **Bank Network**

- Hundreds of routers
- Thousands of switches

All devices send logs to a **central syslog server**.

Benefits:

- Security auditing
 - Incident investigation
 - Faster troubleshooting
-

Interview One-Line Answer

“SYSLOG is an industry-standard protocol used to record and categorize system events using severity levels.”

Final Key Points

- ✓ SYSLOG is used for **logging system events**
 - ✓ Severity levels define **event importance**
 - ✓ Logs can be stored locally or sent to servers
 - ✓ External syslog servers are common in real networks
 - ✓ Works together with **SNMP for monitoring and troubleshooting**
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CHAPTER SUMMARY



What is SYSLOG?

SYSLOG (System Logging Protocol) is an industry-standard protocol used to **record and manage log messages generated by network devices.**

Network devices like:

- Routers
- Switches
- Firewalls
- Servers

generate **log messages when events occur.**

These messages are called **logs.**

SYSLOG records events such as:

- System status
- Errors
- Warnings
- Configuration changes
- Security events



In simple words:

SYSLOG is like a diary of everything happening inside a network device.



Why SYSLOG is Needed?

Imagine a situation:

At **2:00 AM**, a router suddenly stops working.

Without SYSLOG ❌

- Engineers don't know what happened

- Troubleshooting becomes very difficult

With SYSLOG 

Logs show exactly what happened:

- Interface went down
- Routing protocol failed
- Authentication failed
- Device rebooted

 Think of SYSLOG like **CCTV camera for network devices**.

It records every event happening inside the device.

What Events Does SYSLOG Log?

SYSLOG records many important events.

Examples:

- ✓ Interface UP or DOWN
- ✓ OSPF neighbor status change
- ✓ System reboot
- ✓ Configuration changes
- ✓ Login attempts
- ✓ Security violations
- ✓ Hardware failures

These logs help engineers **diagnose problems quickly**.

Where SYSLOG Messages Are Stored

SYSLOG messages can be stored in multiple locations.

Location	Description
Console	Displayed on device console
Buffer	Stored temporarily in RAM

Location	Description
VTY	Displayed in Telnet/SSH sessions

External Server Sent to centralized log server

External syslog servers are the **most common method in enterprise networks**.

🔗 Relationship Between SYSLOG and SNMP

Both protocols help monitor networks but they work differently.

SYSLOG	SNMP
Logs events	Monitors device data
Push-based	Pull-based
Records messages	Retrieves information
Cannot modify config	Can modify variables

📌 Simple explanation:

SYSLOG → Event recording 📄

SNMP → Device monitoring 📊

Both are used together in enterprise networks.

✉️ SYSLOG Message Format

SYSLOG messages follow a structured format.

Example format:

seq: timestamp: %facility-severity-MNEMONIC: description

Each part provides specific information about the event.

Message Fields Explanation

Field	Meaning
-------	---------

seq	Message sequence number
-----	-------------------------

timestamp	Time when event occurred
-----------	--------------------------

facility	Process generating the log
----------	----------------------------

severity	Importance level
----------	------------------

mnemonic	Event code
----------	------------

description	Detailed message
-------------	------------------

What is a Mnemonic?

Mnemonic is a **short code representing an event**.

Example:

LINK-3-UPDOWN

Meaning:

Part	Meaning
------	---------

LINK	Interface event
------	-----------------

3	Severity level
---	----------------

UPDOWN	Interface status changed
--------	--------------------------

SYSLOG Severity Levels

SYSLOG uses **8 severity levels (0–7)**.

Level	Keyword	Meaning
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0	Emergency	System unusable
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1	Alert	Immediate action required
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	Level	Keyword	Meaning
	2	Critical	Critical condition
	3	Error	Error condition
	4	Warning	Warning condition
	5	Notification	Normal but important
	6	Informational	Informational message
	7	Debugging	Debug messages



Easy Memory Trick

To remember severity levels:

Every Awesome Cisco Engineer Will Need Ice Cream Daily 🍦

Letter Meaning

E	Emergency
A	Alert
C	Critical
E	Error
W	Warning
N	Notification
I	Informational
D	Debugging



Real-World Importance

SYSLOG severity levels help companies determine **incident priority**.

Important IT operations concepts:

Concept Meaning

SLA Service Level Agreement

MTTR Mean Time To Resolve

MTR Mean Time To Respond

ITIL Incident management framework

Severity levels help define:

- Issue priority
- Response time
- Escalation level

SYSLOG Logging Locations

Console Logging

Logs appear directly on the device console.

Used during **direct device access**.

Buffer Logging

Logs stored in **device RAM**.

Important:

- Lost after reboot
- Useful for quick troubleshooting

VTY Logging

Logs displayed during **Telnet or SSH sessions**.

Disabled by default.

External Syslog Server

Most common method.

Advantages:

- Centralized logging
- Long-term storage
- Security auditing

 Uses **UDP Port 514**.

Severity Level Behavior

Example configuration:

logging trap warning

Router logs:

- Warning (4)
- Error (3)
- Critical (2)
- Alert (1)
- Emergency (0)

 Logging always works **upwards toward level 0**.

Terminal Monitor

When connected via **Telnet or SSH**, logs are not displayed by default.

Command to enable:

terminal monitor

This must be run **each time a new session starts**.

Logging Synchronous

Problem:

Syslog messages may interrupt CLI commands.

Solution:

logging synchronous

This keeps CLI **clean and readable**.

Timestamp and Sequence Numbers

Commands to improve logs:

Enable timestamps:

service timestamps log datetime msec

Enable sequence numbers:

service sequence-numbers

Benefits:

- Identify event time
 - Identify event order
-

SYSLOG LAB SUMMARY

Enable Buffer Logging

logging buffered 8192

Verify logs:

show logging

Configure External Syslog Server

logging host 192.168.1.100

logging trap debugging

Generate Event

```
interface g0/1  
shutdown
```

Logs appear on the **syslog server**.



CHAPTER CONCLUSION

SYSLOG is a **critical protocol used for logging and monitoring events in network devices**.

Key takeaways:

- ✓ Records system events and errors
- ✓ Uses severity levels (0–7) to categorize events
- ✓ Logs can be stored locally or centrally
- ✓ External syslog servers provide centralized monitoring
- ✓ Works together with SNMP for full network visibility

SYSLOG is widely used in:

- Enterprise networks
- Data centers
- Internet service providers
- Security monitoring systems

Understanding SYSLOG is **essential for network troubleshooting and security analysis**.

Q & A

1 What is SYSLOG?

SYSLOG is an **industry-standard protocol used for recording and managing log messages generated by network devices.**

It allows network administrators to **monitor system events, troubleshoot problems, and audit security activities.**

2 Why is SYSLOG important?

SYSLOG is important because it provides a **record of events occurring on network devices.**

These logs help engineers:

- Detect failures
 - Troubleshoot problems
 - Monitor system health
 - Investigate security incidents
-

3 What types of events does SYSLOG record?

SYSLOG records events such as:

- Interface status changes
 - System reboot
 - Routing protocol failures
 - Login attempts
 - Configuration changes
 - Hardware errors
-

4 Where can SYSLOG messages be stored?

SYSLOG messages can be stored in:

- Console
- Buffer memory
- VTY sessions
- External syslog server

External syslog servers are most commonly used in enterprise networks.

5 What is SYSLOG severity level?

SYSLOG severity levels define the **importance of a log message**.

Levels range from **0 (most critical) to 7 (least critical)**.

6 What port does SYSLOG use?

SYSLOG uses:

- **UDP Port 514**

for transmitting log messages to a syslog server.

7 What is a SYSLOG mnemonic?

A mnemonic is a **short code used to describe the type of event**.

Example:

LINK-3-UPDOWN

This indicates that the **interface link status changed**.

8 Difference Between SYSLOG and SNMP?

SYSLOG

Records events

Push-based

Logs messages

SNMP

Monitors performance

Pull-based

Retrieves data

9 How do you enable syslog buffering?

logging buffered 8192

This command stores logs in device RAM.

10 How do you send logs to an external syslog server?

logging host 192.168.1.100

logging trap debugging

This sends syslog messages to the server.

1 1 What command displays logs?

show logging

Displays all stored syslog messages.

1 2 What does terminal monitor do?

The command:

terminal monitor

enables syslog messages to appear during **Telnet or SSH sessions**.

1 3 Why use logging synchronous?

logging synchronous

prevents syslog messages from interrupting CLI commands.

1 4 Practical Interview Scenario

Question:

Router interface suddenly goes down.
How will you find the reason?

Answer:

Check syslog messages using:

show logging

The logs will indicate the **exact event and timestamp**.
